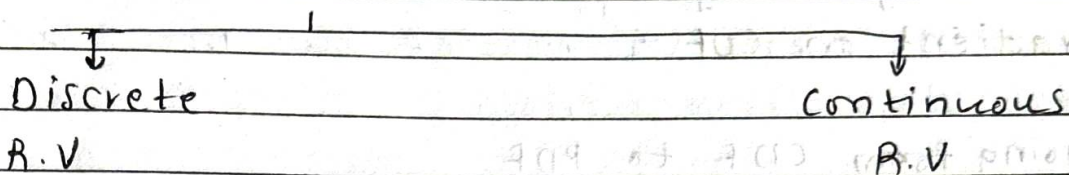


* Random Variable



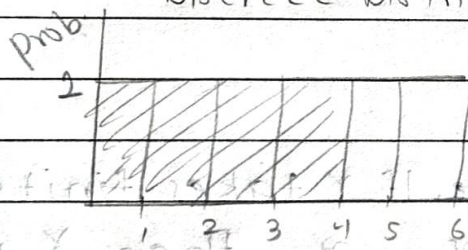
eg: coin toss, dice
- Finite and whole numbers

- infinite within a range
- can be decimals

- Probability Mass function (PMF) - Discrete
- Probability density function (PDF) - Continuous
- Cumulative Distribution Function (CDF)

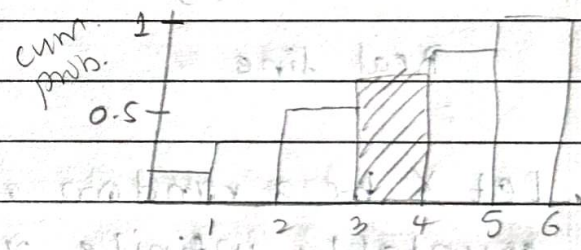
Q. What is PMF, PDF and CDF? What is the relationship between them?

→ Discrete Distribution



(outcome)

PMF

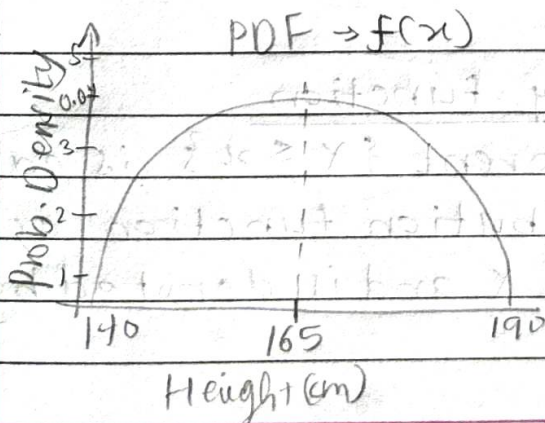


(outcome)

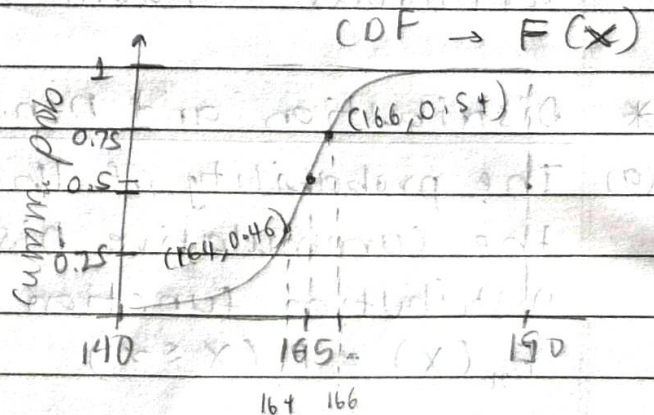
CDF

$$P(X \leq 4) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

→ Continuous Distribution



PDF $\rightarrow f(x)$



CDF $\rightarrow F(x)$

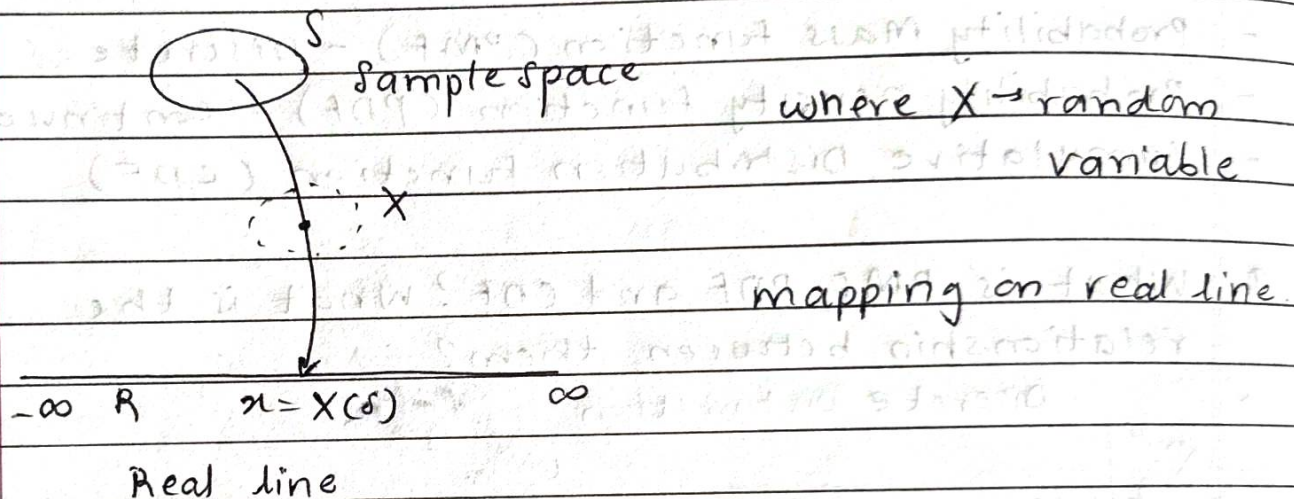
$$\frac{\text{rise}}{\text{run}} = \frac{0.08}{1} = 0.08 \text{ (gradient)}$$

Probability density represents the value of gradient on CDF.

Going from CDF to PDF

$$1. \frac{d}{dx} F(x) = f(x)$$

$$2. \int f(x) dx = F(x)$$



- Let X be a random variable. If X takes finite or countably infinite $x_0, x_1, x_2, \dots, x_n$ then X is called discrete Random variable (DRV)
- Let X be a random variable. If X takes uncountably infinite values in a given interval then X is called continuous Random variable. (CRV)

* Distribution and Density Function

- (a) The probability of the event $\{X \leq x\}$ is called the Cumulative Distribution function or the Distribution function of X and is denoted by $F_x(X) = P(X \leq x)$

(b) If X is a random variable and if $F_X(x)$ is its distribution function then the derivative of $F_X(x)$ wrt x is called Probability Density Function or Density Function.

$$f_X(x) = \frac{d F_X(x)}{dx}$$

Q. Let X be a random variable taking values 1, 2 and 3 with probabilities $\frac{3}{15}$, $\frac{7}{15}$ and $\frac{5}{15}$.

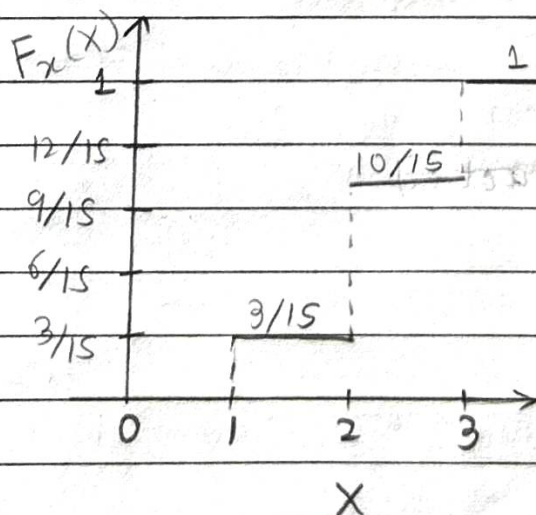
Find its distribution funcⁿ and show the distribution funcⁿ diagrammatically.

$X :$	1	2	3
$P(X=x) :$	$3/15$	$7/15$	$5/15$

$$F_X(X \leq 1) = P(X=1) = \frac{3}{15} = \frac{1}{5}$$

$$F_X(X \leq 2) = P(X=1) + P(X=2) = \frac{3}{15} + \frac{7}{15} = \frac{10}{15} = \frac{2}{3}$$

$$F_X(X \leq 3) = P(X=1) + P(X=2) + P(X=3) = \frac{3}{15} + \frac{7}{15} + \frac{5}{15} = 1$$



- Q. Write the probability density function and also the distribution function of the sum of the numbers appearing when a pair of dice is thrown.

PDF (Probability Density function) -

X:	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$:	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

D1 \ D2	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

CDF

Distribution function

X:	2	3	4	5	6	7	8	9	10	11	12
$P(X \leq x)$:	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$

$P(X=x)$



probability
of success

number of
trials conducted

* Properties of Distribution Function

1. $0 \leq F_X(x) \leq 1$
2. $F_X(-\infty) = 0$
3. $F_X(\infty) = 1$
4. $F_X(x_1) \leq F_X(x_2)$ if $x_1 < x_2$
5. The function is continuous on right
 $F_X(x^+) = F_X(x)$
6. $F_X(x_2) - F_X(x_1) = P(x_1 < X \leq x_2)$
7. $F'(x) = f(x) = \lim_{h \rightarrow 0^+} \frac{P(x \leq X \leq x+h)}{h}$
8. If R.V is discrete then $F_X(x)$ is a step funcⁿ

* Properties of Probability Density Function

1. $f_X(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} f_X(x) dx = 1$
3. $F_X(x) = \int_{-\infty}^x f_X(t) dt$
4. $P(\alpha < X < \beta) = \int_{\alpha}^{\beta} f_X(x) dx$

* Expectation of Random Variable

$$E(X) = \sum p_i x_i, \quad \sum p_i = 1 \rightarrow \text{DRV}$$

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx \quad \text{where} \quad \int_{-\infty}^{\infty} f(x) dx = 1 \rightarrow \text{CRV}$$

Q. A fair coin is tossed three times. A person receives rupees X^2 if he gets X heads. Find his expectation.

$S = HHH, HHT, HTH, THH, TTH, THT, HTT, TTT$

Prob. distⁿ of X :

X	0	1	2	3
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

for X^2

X^2	0	1	4	9
$P(X^2)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$E(X) = \sum p_i x_i$$

$$= \left(\frac{1}{8} \times 0\right) + \left(\frac{3}{8} \times 1\right) + \left(\frac{3}{8} \times 4\right) + \left(\frac{1}{8} \times 9\right)$$

$$= 3 \text{ Rs.}$$

Q. Let X denote the order of the toss at which a fair coin is tossed till a head appears. What is the expectation of the number of tosses required?

Event H TH TTH $TTTH$...

X	1	2	3	4	...
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	...

$$E(X) = \sum p_i x_i$$

$$S = \left(\frac{1}{2} \times 1\right) + \left(\frac{1}{4} \times 2\right) + \left(\frac{1}{8} \times 3\right) + \left(\frac{1}{16} \times 4\right) + \dots$$

$$\frac{1}{2} S = \frac{1}{4} + 2 \cdot \frac{1}{8} + 3 \cdot \frac{1}{16} + \dots$$

$$S - \frac{1}{2} S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$= \frac{1}{2} \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right] \text{ G.P. series } \therefore \frac{a}{1-r}$$

$$\frac{1}{2} S = \frac{1}{2} \left[\frac{1}{1 - (1/2)} \right] = 1$$

$$\therefore S = E(X) = 1 + 1 = 2$$

Q. Find expectation of X if X has the PDF
 $f(x) = \begin{cases} kx(2-x), & 0 \leq x \leq 2, k > 0 \\ 0, & \text{elsewhere} \end{cases}$

→

$$E(X) = \int_0^2 x f(x) dx = \int_0^2 x \cdot kx(2-x) dx$$

$$= k \int_0^2 x^2 (2-x) dx$$

$$= k \int_0^2 (2x^2 - x^3) dx = \frac{4k}{3}$$

- Geometric progression
- Basics of integration
- Integration of exponential terms
- Integration with ∞ , limits

Q. The daily consumption of electric power in million kWh is a random variable X with probability density function. Find the value of k such that

$$f(x) = \begin{cases} kx e^{-x/3} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

k , expectation and the probability that on a given day the electric consumption is more than expected value.

1) $\int_{-\infty}^{\infty} f(x) dx = 1$

$$k \int_0^{\infty} x \cdot e^{-x/3} dx = 1$$

2) $E(X) = \int_0^{\infty} x k x e^{-x/3} dx \quad \dots \int_0^{\infty} x f(x) dx$

$$= k \int_0^{\infty} x^2 \cdot e^{-x/3} dx$$

$$= \frac{1}{9} \int_0^{\infty} x^2 \cdot e^{-x/3} dx = 6$$

$$3) P(X > 6) = \int_6^{\infty} f(x) dx = \frac{1}{9} \int_6^{\infty} x \cdot e^{-x/3} dx$$

$$= 0.406$$

Mean

$$\mu_1' = E(X) = \sum p_i x_i \text{ or } \mu_1' = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\mu_2' = E(X^2) = \sum p_i x_i^2 \text{ or } \mu_2' = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \mu_2' - (\mu_1')^2$$

Q. If X denotes the smaller of the two numbers including equality that appears when a pair of dice is thrown. Find the probability distribution of X . Also find mean and variance of X .

1. Probability Distribution of X

X :	1	2	3	4	5	6
$P(X=x)$:	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$

$$E(X) = \sum p_i x_i$$

$$= \left(1 \times \frac{11}{36}\right) + \left(2 \times \frac{9}{36}\right) + \left(3 \times \frac{7}{36}\right) + \left(4 \times \frac{5}{36}\right) + \left(5 \times \frac{3}{36}\right) + \left(6 \times \frac{1}{36}\right)$$

$$= 2.527$$

$$\begin{aligned}
 E(X^2) &= \sum p_i x_i^2 \\
 &= \left(1 \times \frac{11}{36}\right) + \left(4 \times \frac{9}{36}\right) + \left(9 \times \frac{7}{36}\right) + \left(16 \times \frac{5}{36}\right) \\
 &\quad + \left(25 \times \frac{3}{36}\right) + \left(36 \times \frac{1}{36}\right) \\
 &= 8.36
 \end{aligned}$$

$$\begin{aligned}
 V(X) &= E(X^2) - [E(X)]^2 = 8.36 - (2.527)^2 \\
 &= 1.97
 \end{aligned}$$

0. A discrete random variable has the probability density function given below.

$X :$	-2	-1	0	1	2	3
$P(X=x) :$	0.2	k	0.1	$2k$	0.1	$2k$

find k , mean and variance.

1) $k :$ $\sum p_i = 1$

$$0.2 + k + 0.1 + 2k + 0.1 + 2k = 1$$

$$5k = 1 - 0.4$$

$$5k = 0.6$$

$$k = \frac{0.6}{5}$$

$$k = 0.12$$

2) Mean = $E(X) = \sum p_i x_i$

$$\begin{aligned}
 &= (-2 \times 0.2) + (-1 \times 0.12) + 0 + (1 \times 0.24) \\
 &\quad + (2 \times 0.1) + (3 \times 0.24) \\
 &= 0.64
 \end{aligned}$$

$$\begin{aligned}
 3) \quad E(X^2) &= \sum p_i x_i^2 \\
 &= (4 \times 0.2) + (1 \times 0.12) + (1 \times 0.24) \\
 &\quad + (4 \times 0.1) + (9 \times 0.24) \\
 &= 3.72
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - [E(X)]^2 = 3.72 - (0.64)^2 \\
 &= 3.31
 \end{aligned}$$

Q. A continuous random variable X has PDF defined by

$$f(x) = A + Bx, \quad 0 \leq x \leq 1$$

If mean of the distribution is $1/3$, find A & B .

$$\rightarrow \int_{-\infty}^{\infty} f(x) dx = 1$$

$$E(X) = \int_0^1 x f(x) dx$$

$$\int_0^1 (A + Bx) dx = 1$$

$$\int_0^1 x(A + Bx) dx = \frac{1}{3}$$

$$\left[Ax + Bx^2 \right]_0^1 = 1$$

$$\int_0^1 (Ax + Bx^2) dx = \frac{1}{3}$$

$$\begin{aligned}
 \frac{A}{2} + \frac{B}{3} &= 1 \quad \text{--- (1)} \\
 \left[\frac{Ax^2}{2} + \frac{Bx^3}{3} \right]_0^1 &= \frac{1}{3}
 \end{aligned}$$

$$\frac{A}{2} + \frac{B}{3} = 1 \quad \text{--- (1)}$$

$$\frac{A}{2} + \frac{B}{3} = \frac{1}{3} \quad \text{--- (2)}$$

$\therefore A = 2, B = -2$ after solving (1) and (2) simultaneously.

Q. A continuous random variable X has the PDF $f(x) = kx^2 e^{-x}$, $x \geq 0$. Find k , mean and variance.

$$\int_0^{\infty} kx^2 e^{-x} dx = 1$$

$$k \int_0^{\infty} x^2 \cdot e^{-x} dx = 1$$

$$\int uv = u \cdot \frac{dv}{dx} - u \int \frac{dv}{dx}$$

$$k = \frac{1}{2}$$

$$E(X) = \frac{1}{2} \int_0^{\infty} x^3 e^{-x} dx = 3$$

$$\mu_2' = \frac{1}{2} \int_0^{\infty} x^4 \cdot f(x) dx = \frac{1}{2} \int_0^{\infty} x^4 \cdot e^{-x} dx = 12$$

$$V(X) = \mu_2' - (\mu_1')^2 = 12 - 3^2 = 3$$

Q. A continuous random variable X has the following probability law.

$$f_X(X) = kx^2, \quad 0 \leq x \leq 2$$

Find the normalising factor k and find the probabilities $P(0.2 \leq x \leq 0.5)$ and probability $X \geq \frac{3}{4}$ given that $X \geq \frac{1}{2}$ and obtain

distribution function.

$$1. \int_a^b f(x) dx = 1$$

$$\int_0^2 kx^2 dx = 1$$

$$k \int_0^2 x^2 dx = 1$$

$$k \left[\frac{x^3}{3} \right]_0^2 = 1$$

$$k \frac{8}{3} = 1$$

$$k = \frac{3}{8}$$

$$2. \frac{3}{8} \int_{0.2}^{0.5} x^2 dx = \frac{3}{8} \left[\frac{x^3}{3} \right]_{0.2}^{0.5}$$

$$= \frac{3}{8} \left[\frac{(0.5)^3}{3} - \frac{(0.2)^3}{3} \right]$$

$$= 0.0146$$

$$3. \quad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A) = P(X \geq 1/2)$$

$$P(B) = P(X \geq 3/4)$$

$$P(A \cap B)$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

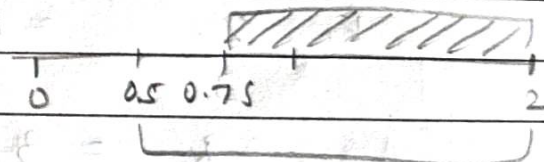
$$P(A) = P(X \geq 1/2)$$

$$= \frac{3}{8} \int_{0.5}^2 x^2 dx = 0.9843$$

$$P(B) = P(X \geq 3/4)$$

$$= \frac{3}{8} \int_{0.75}^2 x^2 dx = 0.947$$

$$P(A \cap B) = 0.947$$



$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.947}{0.9843} = 0.96$$

$$4. \quad F_x(X) = \int_{-\infty}^x f(x) dx = \int_0^x \frac{3}{8} x^2 dx$$

$$= \frac{3}{8} \left[\frac{x^3}{3} \right]_0^x$$

$$= \frac{x^3}{8}$$

Q. Let X be a continuous random variable with p.p.f
 $f(x) = kx(1-x)$, $0 \leq x \leq 1$

Find k and determine number b such that
 $P(X \leq b) = P(X \geq b)$.

Also obtain distribution function

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 kx(1-x) dx = 1$$

$$k \int_0^1 (x - x^2) dx = 1$$

$$k \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

$$k \left[\frac{3}{6} - \frac{2}{6} \right] = 1$$

$$k = 6$$

$$k = 6$$

Total probability = 1

$$P(X \leq b) = \frac{1}{2}$$

$$\rightarrow \int_0^b f(x) dx = \frac{1}{2}$$

$$6 \int_0^b (x - x^2) dx = \frac{1}{2}$$

$$6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^b = \frac{1}{2}$$

$$3b^2 - 2b^3 = \frac{1}{2}$$

$$6b^2 - 4b^3 - 1 = 0 \rightarrow 4b^3 - 6b^2 + 1 = 0$$

$$b = -0.36 (X)$$

$$b = 1.36 (X)$$

$$b = 0.5 (\checkmark)$$

$$F_X(x) = \int_{-\infty}^x f(x) dx \rightarrow \text{general formula.}$$

$$= 6 \int_0^x (x - x^2) dx$$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^x = \frac{3}{2}x^2 - \frac{2}{3}x^3$$

$$= \boxed{3x^2 - 2x^3}$$

↓

distribution function.

Binomial Distribution

$\mu = np \rightarrow$ mean of binomial distribution

$\sigma = \sqrt{npq} \rightarrow$ standard deviation

$$P(X=x) = {}^nC_x \cdot p^x q^{n-x}$$

$$P(X=r) = {}^nC_r \cdot p^r q^{n-r}$$

Assumptions

1. No. of exp conducted are finite 'n'
2. probability of success is 'p' & constant for each exp.

Q A coin is biased so that a head is twice as likely to appear as a tail. IF the coin is tossed six times, find the probability of getting exactly two heads, atleast three heads, atmost three heads

$$H = 2T$$

$$P(H) + P(T) = 1$$

$$2T + T = 1$$

$$T = \frac{1}{3}$$

$$\therefore T = \frac{1}{3}, H = \frac{2}{3} \rightarrow p$$

$$\downarrow$$

$$q$$

(i) exactly 2 heads

 $n = 6$ (six tosses?)

$$P(X=2) = {}^nC_r \cdot p^r \cdot q^{n-r}$$

$$= {}^6C_2 \cdot \left(\frac{2}{3}\right)^2 \cdot \left(\frac{1}{3}\right)^4 = 0.082$$

$$(ii) P(X \geq 3) = P(X=3) + P(X=4) + P(X=5) + P(X=6)$$

$$= 1 - P(X < 3)$$

OR

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 0.899$$

$$(iii) P(X \leq 3) = 0.3194$$

$$= P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

Q. A binomial distribution with parameter $n=5$,
 $8P(X=4) = P(X=2)$ satisfy the property. Find

— (i) p and q — (ii) $P(X \geq 1)$

$$(i) 8 \cdot {}^5C_4 \cdot p^4 \cdot q = {}^5C_2 \cdot p^2 \cdot q^3$$

$$8 (5p^4q) = 10p^2q^3$$

$$\frac{40}{10} = \frac{p^2q^3}{p^4q}$$

$$4 = \frac{q^2}{p^2}$$

$$q^2 = 4p^2$$

$$(1-p)^2 = 4p^2$$

$$1 - 2p + p^2 = 4p^2$$

$$4p^2 - p^2 + 2p - 1 = 0$$

$$3p^2 + 2p - 1 = 0$$

$$p = \frac{1}{3} \quad \therefore q = \frac{2}{3}$$

Q. The mean and variance of binomial variable are 12 and 4 respectively. Find the distribution.

→

$$np = 12 \rightarrow \text{mean}$$

$$npq = 4 \rightarrow \text{variance}$$

$$q = \frac{npq}{np} = \frac{4}{12} = \frac{1}{3}$$

$$\therefore p = \frac{2}{3}$$

$$n \left(\frac{2}{3} \right) = 12$$

$$2n = 36$$

$$n = 18$$

$$\therefore P(X=r) = {}^{18}C_r \cdot \left(\frac{2}{3} \right)^r \cdot \left(\frac{1}{3} \right)^{18-r}$$

Poisson

limiting case of Binomial distⁿ

4

1. Number of trials is indefinitely large $n \rightarrow \infty$
2. Probability P of each trial tends to 0 $P \rightarrow 0$

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

Q. If 5% of electric bulbs manufactured by a company are defective. Find the prob that in a sample of 100 bulbs

- (i) none is defective
(ii) 5 bulbs are defective.

$$p = 5\% = 5/100 = 0.05$$
$$n = 100$$

$$\lambda = np$$

$$(i) P(X=0) = \frac{e^{-\lambda} \cdot \lambda^0}{0!} = 0.006$$

$$\lambda = 100 \times 0.05 = 5$$

$$(ii) P(X=5) = \frac{e^{-5} \cdot 5^5}{5!} = 0.175$$

HW. Q. A manufacturer of blades knows that 5% of his product is defective. If he sells the blades in a box of 100 and guarantees that not more than 10 blades will be defective. What is the probability that a box will fail to meet the guaranteed quality.

Q. Fit the poisson distribution to the following data and calculate theoretical frequency.

x	0	1	2	3	4	Σ
f	123	59	14	3	1	$\Sigma f = 200$
fx	0	59	28	9	4	$\Sigma fx = 100$

Theoretical frequency $\Sigma f \cdot \left[\frac{e^{-\lambda} \cdot \lambda^x}{x!} \right]$

$$\lambda = \frac{\Sigma fx}{\Sigma f} = 0.5$$

Th. freq.	121	61	15	3	0	$\Sigma \text{th. f.} = 200$
-----------	-----	----	----	---	---	------------------------------

* Geometric Distribution Formulae

PMF : $P(X=x) = (1-p)^{x-1} (p)$

CDF : $P(X \leq x) = 1 - (1-p)^x$

Mean : $E(X) = \frac{1}{p}$

Variance : $\text{Var}(X) = \frac{(1-p)}{p^2}$

Standard Deviation : $\sigma = \frac{\sqrt{1-p}}{p}$

- 'until first success is achieved' type of questions.

Theory Questions M2 $(1-x)^9 = (1-x)^9$

Random Variable

- define PMF, PDF, CDF
- explain relation between all three (for TEE)
- define discrete RV, and continuous RV w/ example.
- compare binomial, poisson & geometric distribution.

Q. Suppose that the trainee soldier shoots a target in an independent fashion. If the probability that the target is hit in any shot is 0.7

1. What is the probability that the target would be hit on 10th attempt.
2. Probability that it takes him less than 4 shots.
3. Prob. that it takes him an even num. of shots.

$$1. P(X=10) = q^{x-1} p$$

$$\therefore p = 0.7, q = 0.3$$

$$= (0.3)^9 \cdot 0.7 = 1.377 \times 10^{-5}$$

$$2. P(X < 4) = P(X=1) + P(X=2) + P(X=3)$$

$$= p + q \cdot p + q^2 p$$

$$= (0.973)$$

$$3. P(X = \text{even}) = P(X=2) + P(X=4) + P(X=6) + \dots$$

$$= qp + q^3 p + q^5 p + \dots$$

$$= qp [1 + q^2 + q^4 + \dots]$$

$$= qp \left[\frac{1}{1-q^2} \right]$$

$$= 0.23$$

11

$$4. \quad P(X = \text{odd}) = P(X=1) + P(X=3) + P(X=5) + \dots$$

Q.1. Suppose that the prob. for an applicant for a drivers license to pass a road test on any given attempt is $\frac{2}{3}$. What is the prob. that the applicant will pass the road test on the third attempt.

Q.2. Let X be the number of births in a hospital until the first girl is born. Determine the prob. and distribution function of X . Assume that the probability that the baby born is a girl is 0.5.

Ans. Q.1.

$$p = \frac{2}{3}, q = \frac{1}{3}$$

$$P(X=3) = q^2 \cdot p = \left(\frac{1}{3}\right)^2 \cdot \left(\frac{2}{3}\right) = 0.074 = \frac{2}{27}$$

Ans. Q.2.

$$p = 0.57, q = 0.59 + (0.57)^2$$